

DAY 22 — ADVANCED EVALUATION (SCORING + FORMAT)

GOAL

Upgrade from: raw text scores 

to: structured evaluation  (machines can read + compare)

STEP 1 — FIX EVALUATOR OUTPUT FORMAT : evaluation/evaluator.py

 **FIND THIS:** return result

 **REPLACE WITH:** return self.parse(result)

STEP 2 — ADD PARSER FUNCTION: same file (evaluation/evaluator.py)

ADD BELOW CLASS

```
def parse(self, text):
```

```
    scores = {
```

```
        "relevance": 0,
```

```
        "faithfulness": 0,
```

```
        "correctness": 0
```

```
    }
```

```
    for line in text.split("\n"):
```

```
        if "Relevance" in line:
```

```
            scores["relevance"] = int(line.split(":")[1].strip())
```

```
        elif "Faithfulness" in line:
```

```
            scores["faithfulness"] = int(line.split(":")[1].strip())
```

```
        elif "Correctness" in line:
```

```
            scores["correctness"] = int(line.split(":")[1].strip())
```

```
    return scores
```

✓ ✓ STEP 3 — UPDATE LOGGER FORMAT: evaluation/logger.py

 **FIND:** logs.append(data)

✓ REPLACE WITH

```
data["avg_score"] = (  
    data["evaluation"]["relevance"] +  
    data["evaluation"]["faithfulness"] +  
    data["evaluation"]["correctness"]  
) / 3  
  
logs.append(data)
```

✓ ✓ STEP 4 — PRINT CLEAN SCORES: app/agent.py

 **FIND:** print("\nEVALUATION:\n", evaluation)

✓ REPLACE WITH

```
print("\nEVALUATION:")  
  
print("Relevance:", evaluation["relevance"])  
print("Faithfulness:", evaluation["faithfulness"])  
print("Correctness:", evaluation["correctness"])  
  
avg = (  
    evaluation["relevance"] +  
    evaluation["faithfulness"] +  
    evaluation["correctness"]  
) / 3  
  
print("Average:", avg)
```

✓ ✓ STEP 5 — OPTIONAL (FAIL DETECTION): app/agent.py

✅ ADD AFTER evaluation

if avg < 5:

```
print("[WARNING] Low quality answer")
```

✅ ✅ FINAL FLOW NOW

User Query

↓

Agent

↓

Answer

↓

Evaluator → structured scores ✅

↓

Logger → stores with avg ✅

✅ ✅ TEST

Run: `uvicorn app.api:app --reload`

✅ Output now:

EVALUATION:

Relevance: 8

Faithfulness: 9

Correctness: 8

Average: 8.33

✅ ✅ LOG FILE NOW

logs/history.json

Now stores:

```
{
```

```
"query": "...",  
"answer": "...",  
"evaluation": {  
  "relevance": 8,  
  "faithfulness": 9,  
  "correctness": 8  
},  
"avg_score": 8.33  
}
```

✅✅ **WHAT YOU BUILT:** Structured evaluation system

✅✅ **SYSTEM LEVEL NOW**

- ✅ AI system
 - ✅ API + Docker
 - ✅ Async + caching + streaming
 - ✅ Memory (vector)
 - ✅ Monitoring
 - ✅✅ Advanced evaluation (NEW)
-

✅✅ **WHY THIS IS BIG**

Now you can:

- compare answers
 - track quality improvement
 - detect bad responses
 - build benchmark
-

✅✅ **INDUSTRY TERM**

LLM Evaluation Pipeline

Used in:

- OpenAI evals
 - LangSmith
 - production AI teams
-

✅✅ **NEXT**

Day 23 → Multi-agent system

If ready: 👉 say **Day 23**